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Cave and Landscape Chronology at Timor Caves, New South Wales

R. A. L. OSBORNE

ABSTRACT

Caves have existed at Timor Caves since before the Late Cretaceous and much of the present landscape of the area had its origins prior to the extrusion of Eocene basalts and tuffs.

Main Cave (TR 1) at Timor Caves was formed by the excavation of both bedrock and palaeokarst sediment below the water table under nothephreatic conditions. The palaeokarst sediment represents an extensive former cave filled with speleothem deposited under vadose conditions.

A tabular basalt body exposed in Main Cave is interpreted here as a flow filling a cave at a time intermediate between deposition of the palaeokarst and excavation of the present cave. The basalt has been dated at 73.5 Ma. This provides a maximum age for the present cave void and a minimum age for the palaeokarst.

Two similar basalt bodies exposed in Belfry Cave (TR 2) are also interpreted as flows.

The excavation of Main Cave corresponds with a significant and sustained rise in the local water table most likely associated with filling of the adjacent valley of Isaacs Creek by Eocene (circa 53 Ma) basalt of the Liverpool Range Beds.

Geomorphic evidence suggests that Isaacs Creek adjacent to the caves had a well developed valley before the Eocene basalt was extruded. Since the flows in Belfry Cave did not interact with groundwater, topographic relief during the Late Cretaceous was much the same as at present.

INTRODUCTION

Timor Caves have developed in the Timor Limestone Member of the Middle Devonian Yarrimie Formation (Crook, 1961a) which crops out along the valley of the Isis River, 25 km east of Murrurundi, in the Hunter Valley Region of New South Wales (Fig. 1).

The assignment of a Cretaceous age to Main Cave at Timor by Connolly and Francis (1979) represented a major change in thought about the likely age of caves and karst landscapes in eastern Australia, as previous workers had considred caves to be largely Pleistocene features.

Connolly has more recently retracted from this position (Connolly, 1983), suggesting that the caves must be younger than the Eocene basalts of the Mount Royal Ranges that form the watershed for Isaacs Creek. In a brief comment Francis and Osborne (1983) questioned Connolly's new conclusions and argued that Main Cave had more ancient origins.

Timor Caves have particular potential for studies in speleochronology as they occur close to basalts and intersect tabular basalt bodies. This offers the possibility of dating surface landforms by their relationships with the basalts and also of assigning maximum ages for speleogenetic events through dating of the basalts exposed in the caves.

This paper presents a new view of cave and landscape chronology at Timor Caves based on observations in Main and Belfry Caves, dating of basalt in Main Cave, and consideration of the relationships between the surface landscapes and basalts of Late Cretaceous and Eocene age.

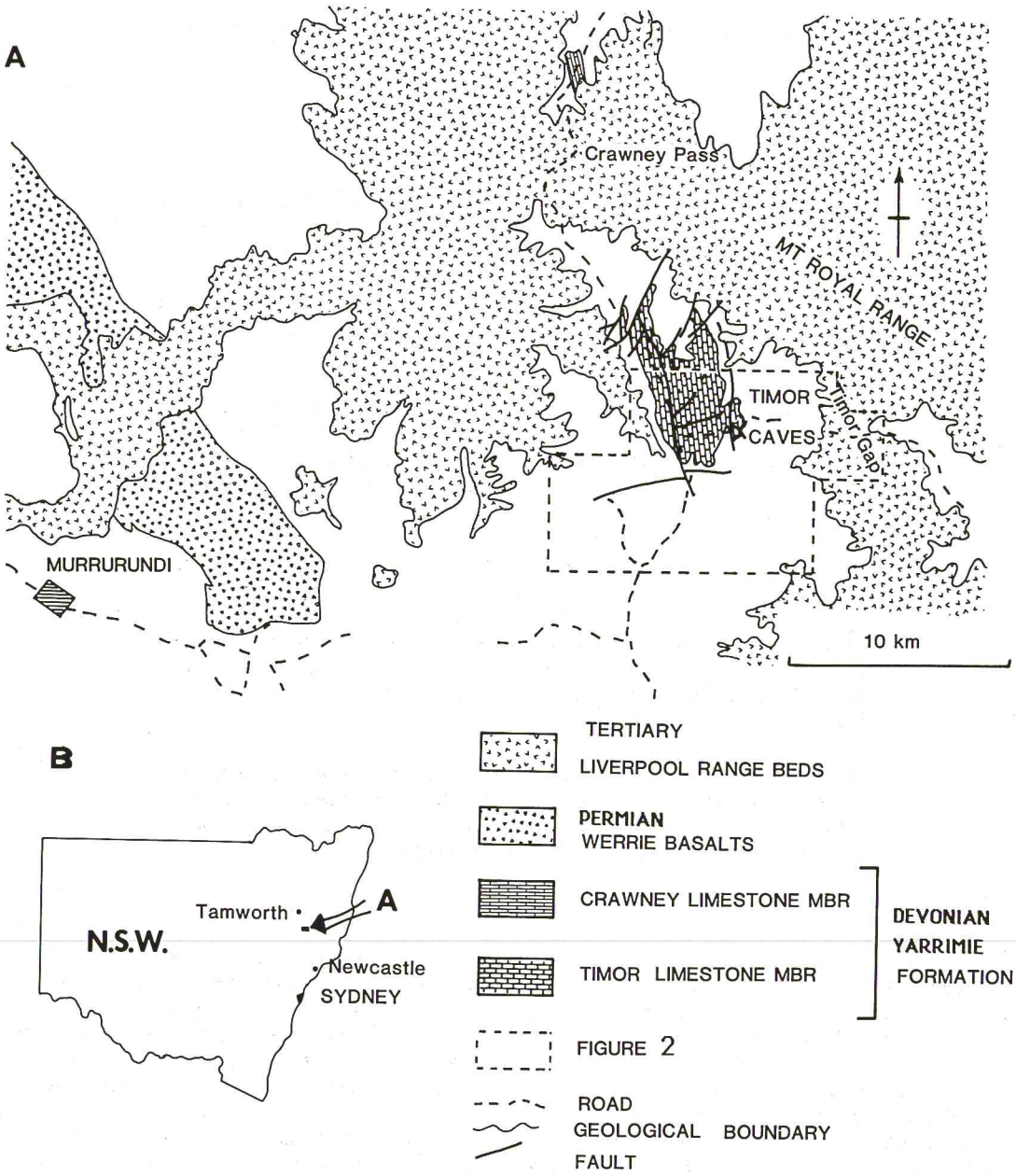


Figure 1. A: Geological setting. Abridged geology after Offenberg (1971).
B: Location

TIMOR CAVES

Forty five caves are recorded at Timor Caves of which thirty are described by James *et. al.* (1976) and a further ten by Warden *et. al.* (1981). Main Cave (TR 1), Belfry Cave (TR 2), and Helictite Cave (TR 4) were described in detail by Connolly and Francis (1979).

Main Cave

Main Cave lies close to the surface high in Caves Ridge (Fig. 3) with an entrance elevation of approximately 640 m A.S.L. It consists (Fig. 4) of a large room, the Ballroom, around which a number of smaller side chambers are developed. Wall and ceiling morphology is indicative of solution under sluggish phreatic (nothephreatic) conditions. The floor is largely composed of grey earth (entrance facies) and guano. Small descending passages at the sides of the Ballroom, developed between the sediment and the walls, suggest that this deposit is quite thick.

Belfry Cave

Belfry Cave is located much lower in Caves Ridge than Main Cave and has its entrance approximately 90 m below that of Main Cave. The cave (Fig. 5) is much more complex than Main Cave and consists of a series of structurally-controlled passages.

The cave can be divided into two quite distinct sections separated by a squeeze over flowstone (A in Fig. 5). The outer section, in which the basalts are exposed, has planed ceilings, channel grooves and asymmetrical wall scallops which Connolly and Francis (1979) recognised as indicating inflow, and at times outflow, through the present entrance, and suggest that this section of the cave was developed (or considerably modified) by shallow phreatic (epiphreatic) and possibly vadose solution. The floor of the outer section is composed of grey earth (entrance facies) similar to that in Main Cave.

The inner section of the cave is significantly structurally controlled and has many features; roof pendants, blades, and spongework which Connolly and Francis (1979) recognised as indicating nothephreatic solution. The atmosphere in the inner section is considerably more humid than that in the outer section and the floor is composed of phosphatic clay.

EVIDENCE FOR MULTIPLE KARSTIFICATION

The term multiple karstification (Osborne, 1984a) is used to describe karst processes affecting the same body of rock on a number of distinct occasions during its history. Multiple karstification results in complex interrelationships between modern karst forms and palaeokarsts of varying ages. Multiple karstification is indicated by the truncation and exposure of palaeokarst sediments in caves.

Connolly and Francis (1979) recognised that Main Cave contains "evidence of substantial former infill" and described how pool crystals and flowstone are exposed in the cave walls. When outlining evidence for multiple karstification in the Lachlan Fold Belt, Osborne (1984a) cited Main Cave at Isaacs Creek as a possible example of multiple karstification in the New England Fold Belt.

Main Cave is an extraordinary example of multiple karstification. Large parts of the cave are developed in flowstone, rather than bedrock. Like the bedrock walls, the walls composed of flowstone (Fig. 6) have been dissolved by nothephreatic solution with wall morphologies being continuous across the unconformity between bedrock and flowstone.

The palaeokarst sediments are composed mainly of flowstone and other speleothem material (both stalactites and stalagmites can be recognised in some exposures, Fig. 7) indicating that the former cave was subjected to vadose conditions for a considerable period after which it re-entered the phreatic zone and the present cave was excavated.

Multiple karstification, although probably on a smaller scale, is indicated by the development of phreatic roof pendants in cemented cave sediments in the inner section of Belfry Cave. These were described and discussed by Connolly and Francis (1979).

Another, apparently older, palaeokarst deposit is exposed in the inner reaches of Belfry Cave (locality "B", Fig. 5) where a body of chert with highly irregular relationships to bedding and the present cave walls is exposed. The chert U.S.G.D. 63251 shows some signs that it may represent a silicified basic igneous rock.

BASALT BODIES IN MAIN AND BELFRY CAVES

Connolly and Francis (1979) reported the presence of amygdaloidal basalt dykes with flow banding in Main and Belfry Caves. They considered that the basalt was too weathered for dating or petrographic study but that, if possible, dating the dykes would provide a maximum age for the caves. Connolly and Francis also proposed that the basalt need not be Tertiary in age but could be Mesozoic since Dulhunty (1976) had established Mesozoic ages for intrusions in the Hunter Valley formerly considered to be of Tertiary age. G. Francis (pers. comm.) suggested that the basalt in the caves might correlate with the Permian Werrie Basalts (see Fig. 2)

In contrast Connolly (1983) concluded that the dykes were the same age as the adjoining basalts of the Mt Royal Ranges thus making the caves less than 50 Ma old.

Francis and Osborne (1983) suggested that although the basalts in the caves are weathered it might still be possible to make petrographic studies of them.

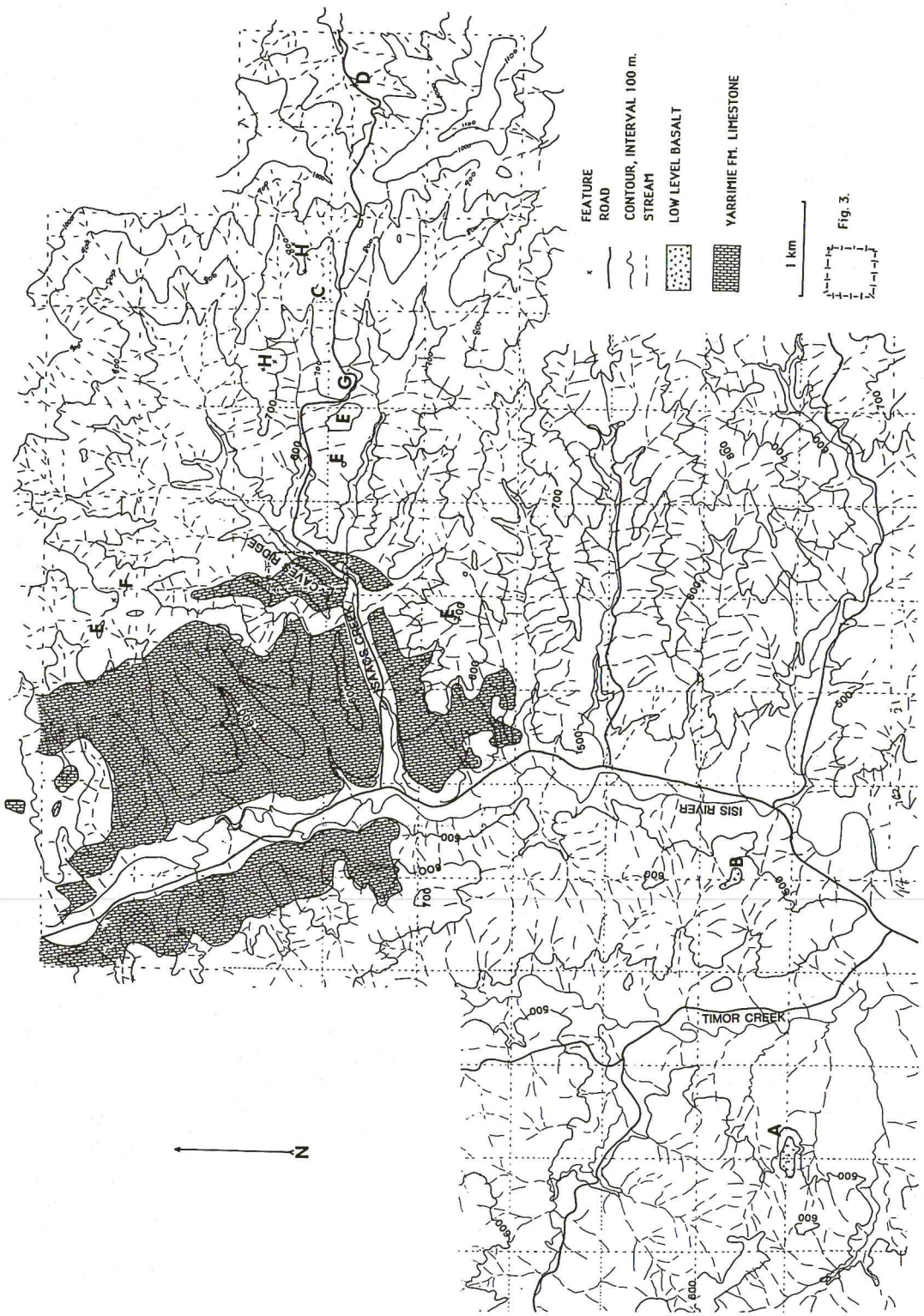

FEATURE
ROAD
CONTOUR, INTERVAL 100 m.
STREAM
LOW LEVEL BASALT
YARRIMIE FM. LIMESTONE
1 km
Fig. 3.
ISIS RIVER
TIMOR CREEK
CATTLE RIDGE
SAWPIT CREEK
A
B
C
D
E
F
G
H
N

Figure 2. Topography of the Timor Caves area. Contours after C.M.A. Isis River 9134-IV-N 1:25 000 Sheet.
A: Low level basalt (U.S.G.D. 62361) mapped by Robson (1982).
B: Low level basalt (U.S.G.D. 62360) mapped by Robson (1982).
C: Tuff (U.S.G.D. 62362) illustrated in Fig. 17.
D: Lowest exposure of basalt (830 m) on road east of Timor Gap.
E: Basalt-free hills with elevations over 700 m A.S.L.
F: Lowest unambiguously in-situ outcrop of basalt on Caves Ridge (circa 790 m).
G: Basalt knoll (U.S.G.D. 62357)
H: Basalt knolls in adition to "G" identified by Connolly and Francis (1979) as having associated magnetic anomolies.
I: Basalt knoll (U.S.G.D. 62356)

Examination of basalts in Main and Belfry caves has shown that they are not as weathered as previous workers had implied and thin sections have been successfully produced from them. The field relationships are more equivocal than has been previously assumed; the decision as to whether the bodies are dykes or flows filling caves is far from simple. It is now possible to compare petrographically the basalts from the caves with the Tertiary basalts. Connolly (1983) has correlated these and with the low level flows in the Isis River Valley described by Robson (1982).

Field Relationships

Three basalt bodies (one in Main Cave and two in Belfry Cave) are truncated by development of the present caves and thus, as recognised by both Connolly and Francis (1979) and Connolly (1983), set a maximum age for the development of the present caves. This, however, does not rule out the possibility that they may be flows filling former caves rather than dykes.

The basalt in Main Cave is exposed in horizontal cross-section in the cave roof and in vertical cross-section in a niche in the south western corner of The Ballroom.

Where it is exposed in vertical cross-section the basalt does not extend to the cave floor, but rather, overlies a palaeokarst deposit. The basalt (Fig 8) is bounded on two sides by massive limestone walls which at the base of the exposure enclose palaeokarst deposits and some 2 m above the cave floor enclose the basalt. The lateral boundary between the basalt and the limestone is sinuous, as is the trace of the boundary in the cave roof.

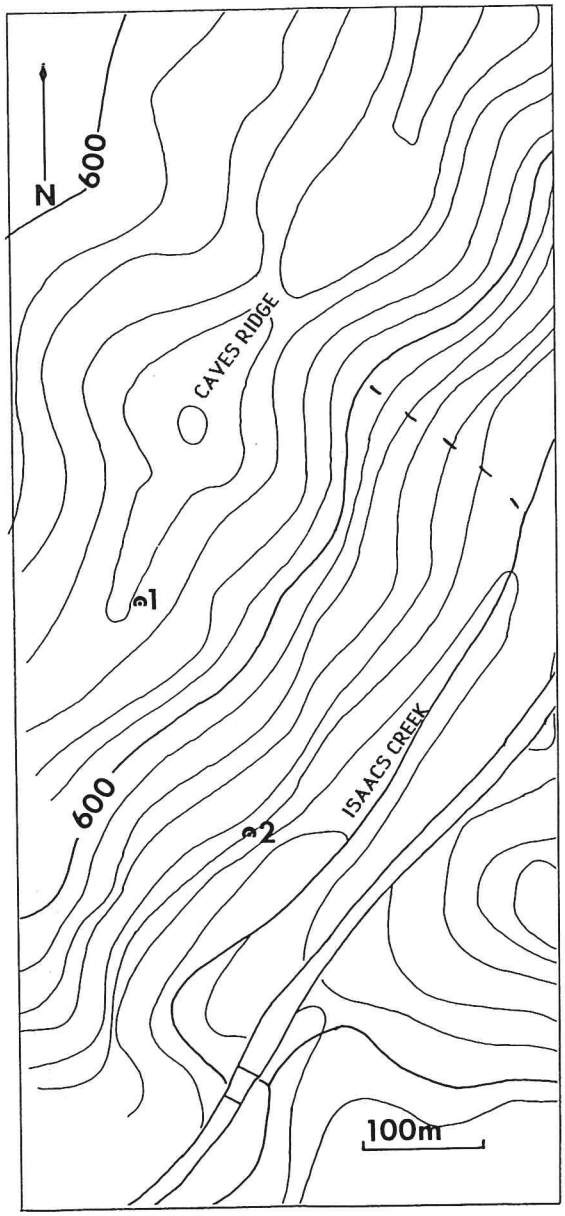


Figure 3. Location of Main and Belfry Caves. Contour base after C.M.A. Isis River 9134-IV-N 1:25 000 Sheet, cave entrance positions after James *et. al.* (1976).

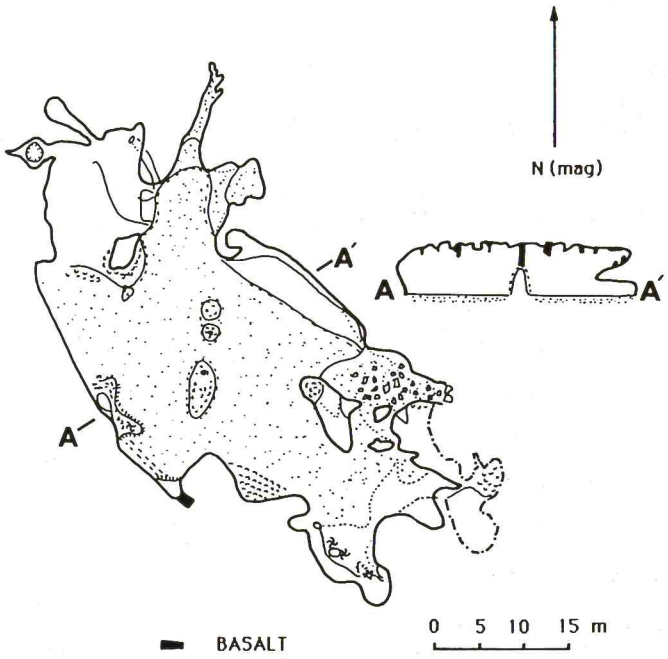


Figure 4. Main Cave, TR 1, map (plan view and section A-A') after James *et. al.* (1976).

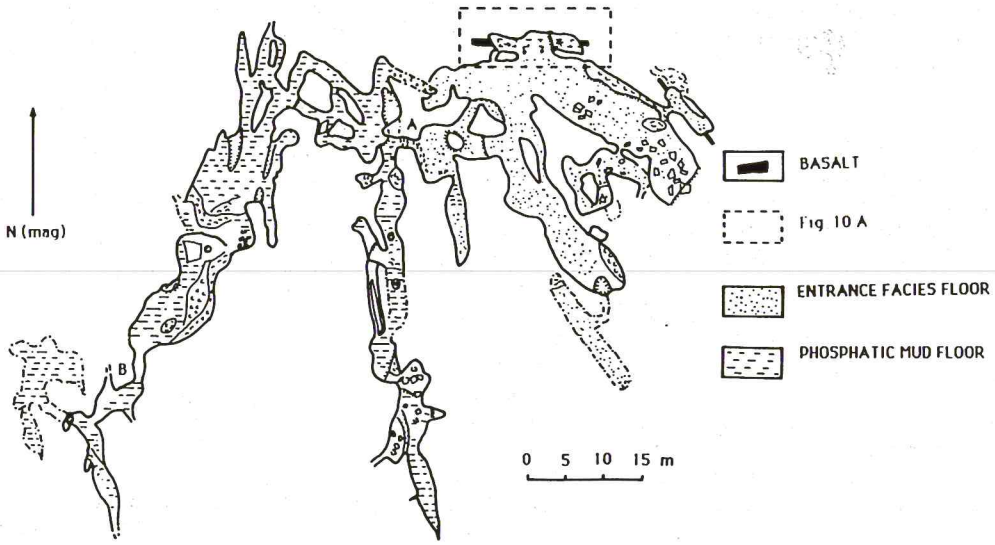


Figure 5. Belfry Cave, TR 2, map (plan view) after Connolly and Francis (1979).
A: Squeeze separating dry outer portion of cave from moist inner portion with phosphatic clay floor.
B: Location of chert body, U.S.G.D.62351.

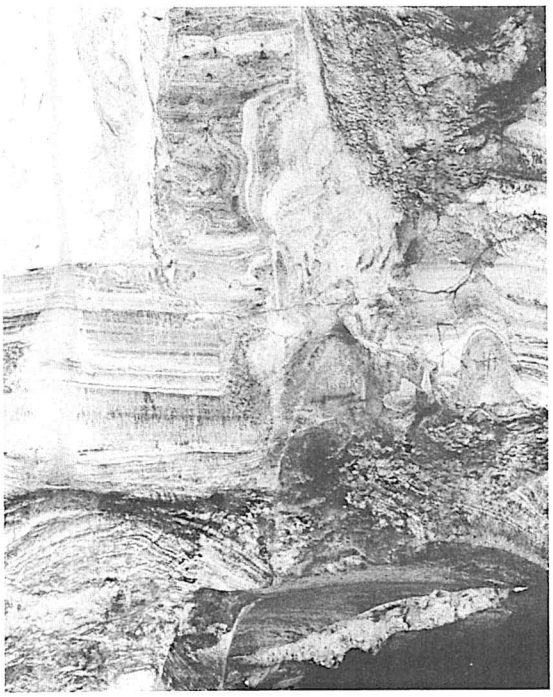


Figure 6.

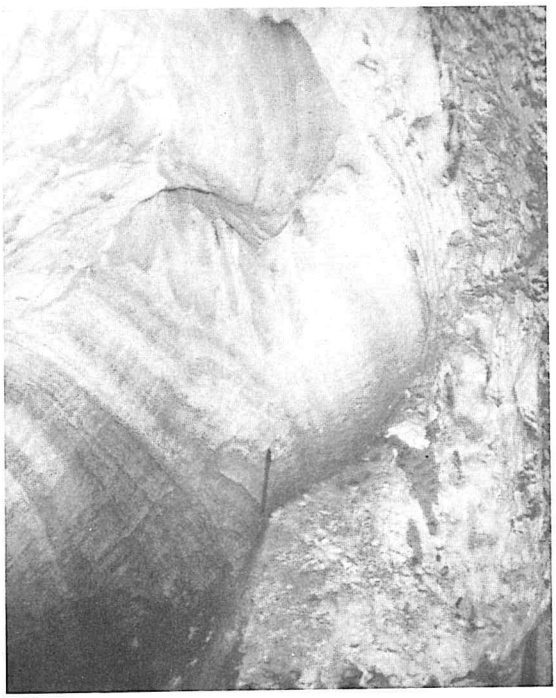


Figure 7.

Figure 6. Palaeokarst exposed in Main Cave. Note how phreatic wall features of present cave continue over the exposed boundaries between bedrock and vadose palaeokarst.

Figure 7. Palaeokarst exposed in Main Cave. In this exposure figured by Osborne (1984b) the unconformable boundaries between bedrock and palaeokarst speleothem and between palaeokarst speleothem and recent speleothem (marked by pen) are visible. Note the preservation of small stalactites within enclosing speleothem in centre of frame.

The observed relationships could have been produced in several ways the two most probable of which are illustrated in Fig. 9. The first, and preferred, history (Fig. 9, A) interprets the basalt as a flow, filling a narrow vadose shaft that had already been partially filled with palaeokarst sediment. With this interpretation the palaeokarst predates the basalt and the present cave is younger than both.

In the second history (Fig. 9, B) basalt intrudes the limestone as a dyke and is partially eroded to form a cave which is later filled by the palaeokarst sediment. The present cave then forms, exposing both the basalt and the palaeokarst. With this interpretation the palaeokarst is younger than the basalt.

The nature of the boundary between the basalt and the limestone supports the former, rather than the latter, interpretation. If the basalt were a dyke emplaced along a joint , as proposed by Connolly and Francis (1979) , it would be expected to have more planar lateral boundaries and very little linear sinuosity. The lateral boundaries between the basalt and the limestone, and the palaeokarst and the limestone, indicate that the cavity in which the basalt and the palaeokarst is enclosed has the form of a vadose cave passage with channel incuts (Fig. 8, A & B).

While the bodies in Belfry Cave are more dyke-like, the western body has parallel walls in the upper part of its exposure but bifurcates downwards (Fig. 10) with an inclined branch sloping towards the main passage of Belfry Cave suggesting that the basalt came from above as a flow entering a cavity, rather than as an intrusion from below. If this body is a flow then it entered a cave above the local water-table, as it shows no sign of interaction with standing water.

Petrography

The igneous bodies in both Main and Belfry Caves are composed of weathered vesicular basalt with flow banding.

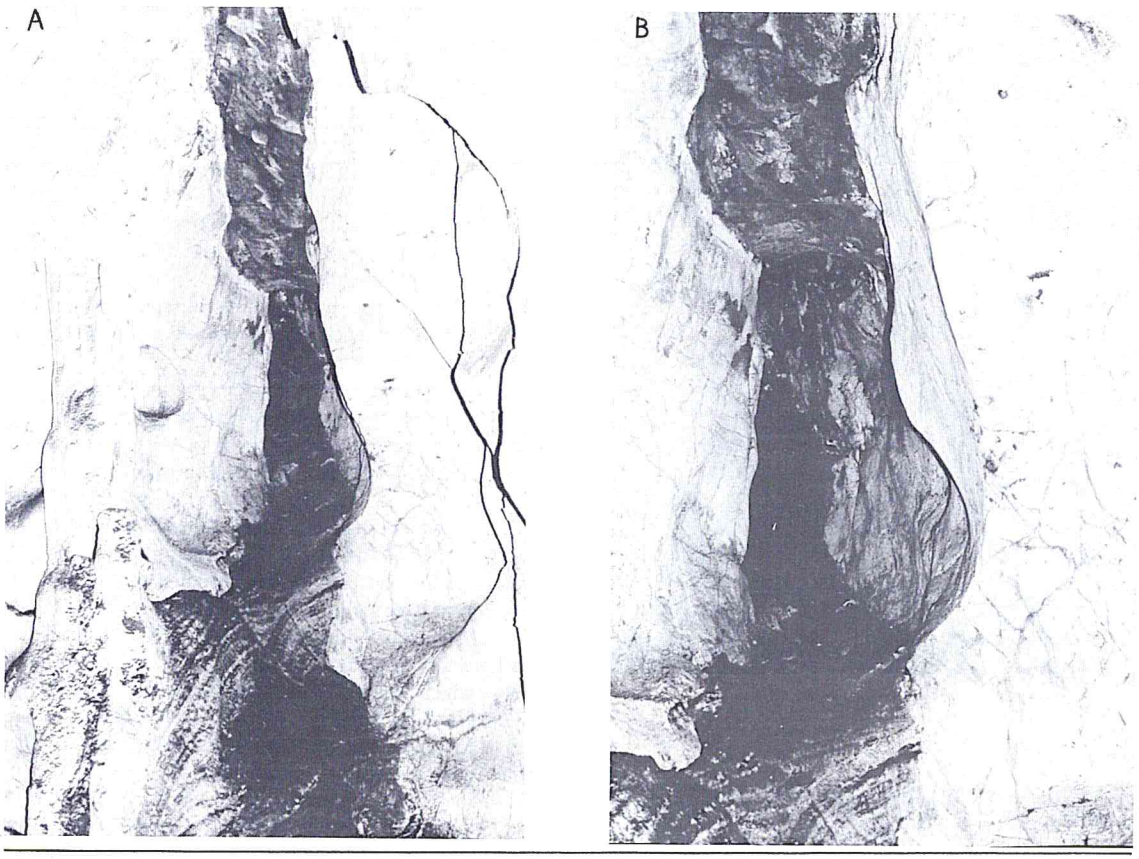


TABLE 1

Potassium-argon data on plagioclase separated from basalt in Timor Caves, 8 km Northeast of Timor, New South Wales.

Sample No.	K wt%	Radiogenic ^{40}Ar (10^{-11} mol/g)	100 Rad. ^{40}Ar / Total ^{40}Ar	Calculated Age (Ma) ± 1 s.d.
84-451	0.291, 0.292	3.82	66.6	73.9 ± 0.8
U.S.G.D. 62359		3.78	73.8	73.2 ± 0.8

$\lambda_e + \lambda_e' = 0.581 \times 10^{-10}\ yr^{-1}$ $\lambda_\beta = 4.962 \times 10^{-10}\ yr^{-1}$

$^{40}K/K = 1.167 \times 10^{-4}$ mol/mol Locality: 31° 42.2' S, 151° 7.2' E.



Figure 8. Basalt body exposed in Main Cave.

A: View of basalt body showing basalt at top and vadose palaeokarst at base, both enclosed by limestone. Note shape of boundary between basalt and limestone.

B: Closer view of basalt/limestone boundary than in "A". Note how vadose channel incut in base of passage is reflected in basalt/limestone boundary and how meandering morphology, typical of vadose cave passages, is shown by basalt/limestone contact in roof.

C: Close up view of basalt at its boundary with palaeokarst, marked by pick, again note shape of basalt-limestone boundary.

Sample U.S.G.D. 62359 from Main Cave (Fig. 12) contains zoned laths of plagioclase (up to 1 mm long) and vesicles filled with banded chalcedony and secondary carbonate. The plagioclase laths have remained relatively intact despite the considerable alteration of much of the rest of the basalt.

A less weathered sample from Belfry Cave, U.S.G.D. 62354, (Fig. 13) again contains plagioclase laths with olivines weathered to clay and carbonate-filled vesicles. A flow texture and carbonate-filled linear structures parallel with the walls of the body (? cooling joints) are developed.

The basalt in the caves is petrographically dissimilar to samples from the ridges to the east, which are titanaugite-rich microgabbros (Fig. 14), but is similar to the plagioclase-rich low-level basalt (B in Fig. 2) mapped by Robson (1982) (Fig. 15). Robson's other low level basalt (A in Fig 2) is in outcrop and petrography more like those to the east of Timor Caves and contains large pyroxene xenocrysts.

Radiometric Dating

The relatively unalterered nature of the plagioclase laths in the basalt from Main Cave (U.S.G.D. 62359) suggested that it might be possible to obtain a mineral date on an extract of the laths. To this end a concentrate of non-magnetic fines consisting largely of plagioclase from the basalt was submitted to the Research School of Earth Sciences, Australian National University for K-Ar dating. The concentrate was purified and the dating carried out with measurements being made in duplicate (Table 1). A date of 73.5 +- 0.8 Ma (Late Cretaceous) was obtained.

A Late Cretaceous date for the basalt in Main Cave may seem anomalous in view of the proximity of extensive Tertiary basalts. There is, however, some evidence for vulcanism of similar age near by.

McDougall and Wilkinson (1967) obtained a date of 58.6 Ma for the Square Top intrustion near Nundle, 25 km north of Timor Caves. They considered this to be a reliable minimum age.

Hollis and Sutherland (1985) have obtained fission track dates of 60.4 Ma for zircons from Duncans Creek east of Nundle and note that this closely matches a recalculated minimum age of 60.1 Ma for the Square Top intrusion. F.L. Sutherland (pers. comm) has found zircons of similar age (60 Ma) in the Gloucester area, 96

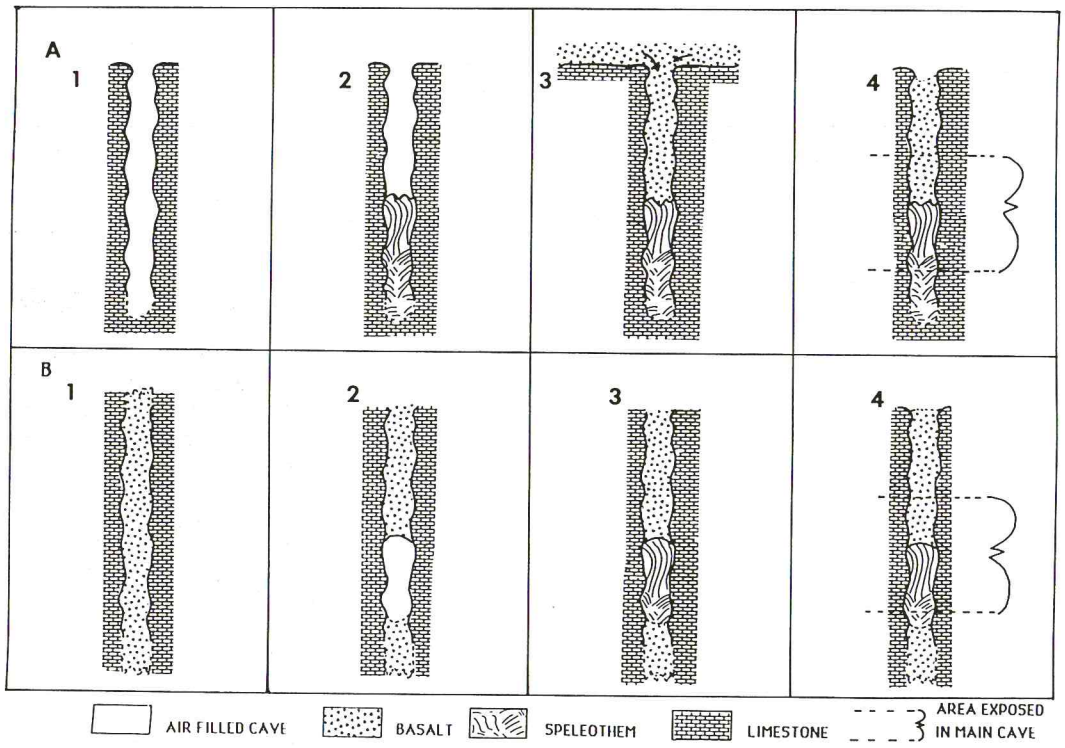


Figure 9. Possible modes of emplacement of basalt body in Main Cave.

A: 1, Formation of narrow cave with connection to surface.
2, Deposition of speleothem in cave.
3, Basalt enters cave from surface flow overlying speleothem.
4, Section between dotted lines is exposed by development of present Main Cave.

B: 1, Basalt dyke intrudes limestone
2, Cave develops by erosion of dyke
3, Cave is filled with speleothem
4, Section between dotted lines is exposed by development of present Main Cave

km east of Timor Caves, suggesting that igneous activity older than the Eocene basalts may have occurred over the same broad area.

Dating of the Liverpool Range Beds as Eocene is based on samples that may represent only the youngest flows in the sequence. Mt Wombramurra, from which the samples dated at 53 Ma by Wellman and Mc Dougall (1974) were obtained, has an elevation of 1417 m, over 600 m above the base of the basalt near Timor Caves. Although, as evidence discussed below suggests, this difference in elevation may not represent the true thickness of the Liverpool Range Beds due to extrusion over an irregular surface (Crook, 1961b considered that their thickness exceeded 300 m). Despite the thickness of the Liverpool Range Beds it seems unlikely that they would range in age over 20 Ma.

The palaeotopography of the Timor Caves area proposed below would require the Late Cretaceous basalts to have flowed from the east or north-east, i.e. from the same direction as the Eocene basalts. It is thus possible that the Late Cretaceous vulcanism represents an earlier phase of activity (evidence for which is mostly now buried) from similar sources to those producing the Liverpool Range Beds.

BASALT AND LANDSCAPE DEVELOPMENT

The Isis River and its upstream tributaries, like Isaacs Creek, rise among the basalt ridges and plateaux of the Mount Royal and Liverpool Ranges.

Crook (1961b) defined the Liverpool Range Beds as a sequence of sub-basaltic sediments and boles together with basalts. These extend from Mt Snowden in the north-west to south and east of Crawney Pass, with representative sections at Snowden Creek and Mt Yellow Rock (19 km north of Isaacs Creek).

Wellman and McDougall (1974) include these basalts as part of their Barrington Volcano and obtained a K/Ar age of 53 Ma for flows at Mt Wombramurra, 7 km north of Isaacs Creek.

Robson (1982) correlated basalt flows in the Yarrabin district with the Liverpool Range Beds.

The Liverpool Range Beds (Crook, 1961b), despite their extension over the Liverpool Range south of Murrurundi Gap by Pogson (1972), are not to be confused with the Liverpool Range Volcano of Wellman and McDougall (1974) or the basalts of the Liverpool Range described by Schön (1982). These form the basalt plateaux and ridges of the Liverpool Range west of Murrurundi and are distinct in age, origin, and composition from the Liverpool Range Beds.

The Liverpool Range Beds unconformably overlie the Devonian Tamworth Group, their basal unconformity being a significant indicator of the early Tertiary landscape. D.F. Branagan (pers. comm.) believes that the sub-basaltic sediments ascribed to the Liverpool Range Beds are significantly older than the basalt.

The Sub-Basaltic Surface

The nature of the sub-basaltic surface is crucial to any discussion of cave and landscape chronology in the Isaacs Creek area. If the sub-basaltic surface is, or was, level and has been disrupted by significant post-basaltic faulting, then caves and landscapes at Isaacs Creek must have originated from erosive events in post-basaltic times. If, however, the sub-basaltic surface had significant relief then the landscapes and older caves could predate the basalt, being covered and later exhumed.

Benson (1913) concluded that the prebasaltic surface was mature in character and trenched by immature valleys. He believed that this supported the idea of Andrews (1911) that peneplanation, uplift, and partial dissection had occurred prior to the eruption of the basalts.

Osborne *et.al.* (1948) recognised a sub-basaltic surface with an average elevation of 2,700-2,800 feet (823-854 m A.S.L.) with pulsatory uplift of the basalt-covered surface resulting in the development of valley-in-valley topography. They also noted "remarkable adjustment" between geological structure and detailed physiographic development of the area.

Crook (1961b) noted the presence of an inlier of Palaeozoic rocks within the Liverpool Range Beds on Ryans Oakey Creek suggesting an uneven prebasaltic surface. He found the Liverpool Range Beds to be tectonically little disturbed and no evidence for post-basaltic faulting.

Galloway (1967) constructed a model of sub-basaltic surfaces in the Hunter Valley in which he assumed that small basalt occurrences are the remnants of a former continuous sheet, that considerable post-basaltic tectonism had produced a sub-basaltic surface which was distinct from the prebasaltic land surface, and that post-basaltic drainage evolved *ab-initio* on a relatively featureless basalt surface.

For the Isaacs Creek area Galloway proposed that the sub-basaltic surface had an elevation of between 2,500 and 2,750 feet (762-838 m A.S.L.).

Connolly and Francis (1979) based their arguments for the antiquity of caves and landscapes on a model of the sub-basaltic surface similar to that of Benson (1913) and Crook (1961b). They proposed a sub-basaltic surface qualitatively like the present surface, with prebasaltic Isaacs Creek flowing in much the same path as the present Isaacs Creek and Main Cave (entrance elevation about 640 m A.S.L.) already well above creek level in prebasaltic Caves Ridge. They further proposed that the basalt filled Isaacs Creek valley and post-basaltic erosion had removed most of the basalt and only slightly deepened the valley into bedrock.

Connolly (1983) rejected this model and returned to the ideas of Galloway (1967) by having Caves Ridge

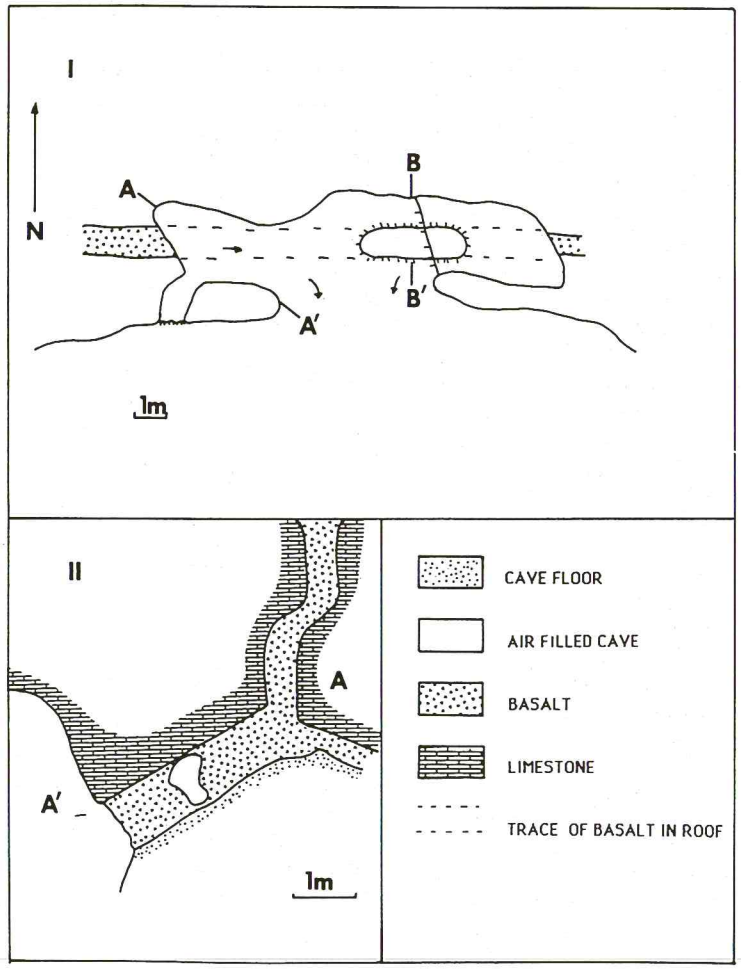


Figure 10. Western basalt body in Belfry Cave
I: Map (plan view) showing basalt body and location of sectional views.
II: Sketch of basalt body viewed in section A-A' (vertical section) showing bifurcation.



Figure 11. Photograph of vertical exposure of western basalt body in Belfry Cave. View corresponds to section line B-B' in Figure. 10, A.

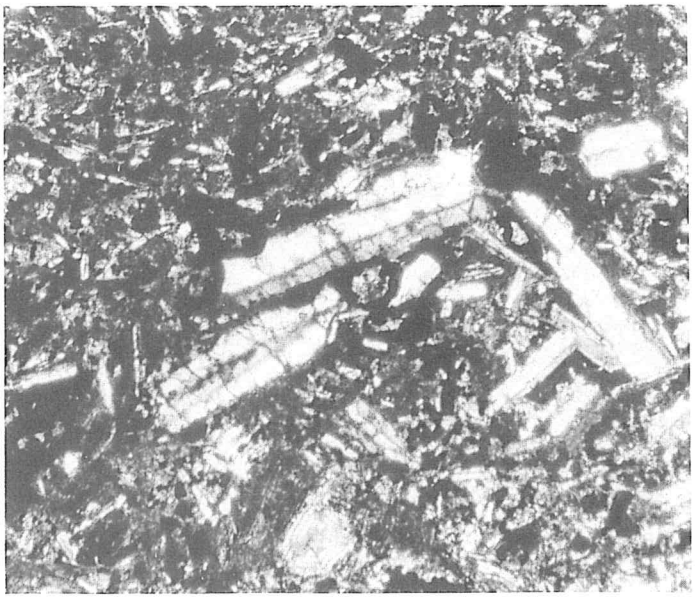


Figure 12. Basalt from Main Cave, U.S.G.D. 62359. crossed nicols, approx. 75 x. Note plagioclase laths and chalcedony-filled vesicle in lower middle of photograph.

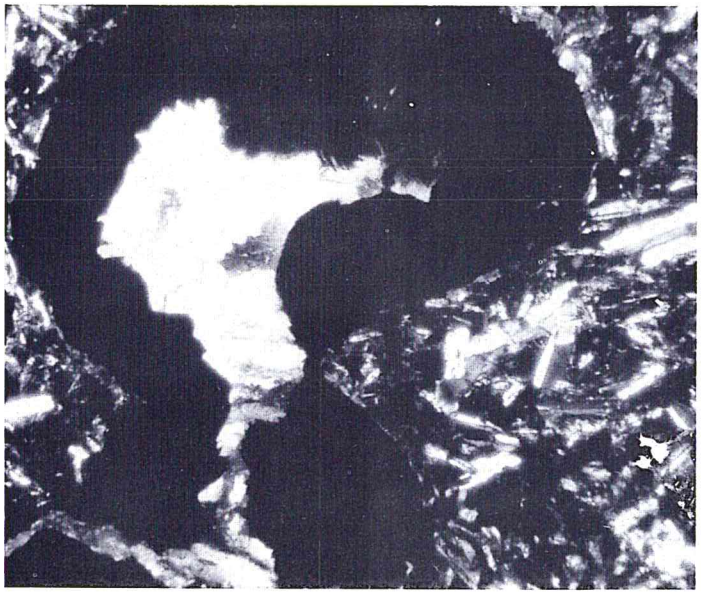


Figure 13. Basalt from Belfry Cave, U.S.G.D. 62354. crossed nicols, approx. 75 x showing calcite and clay filling vesicle.



Figure. 14 Basalt from knoll at "G" in Figure. 2, U.S.G.D. 62357. plane polarized light, approx. 75 x. Note presence of titanaugite and fresh nature of rock.

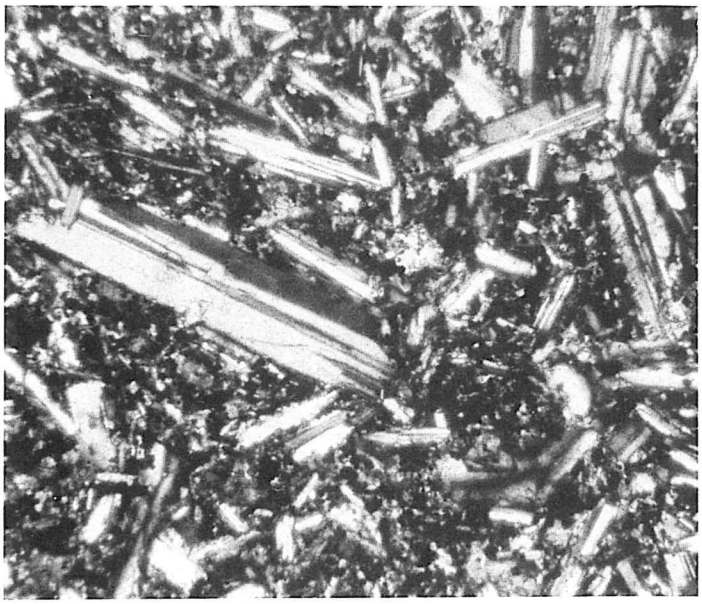


Figure 15. Basalt from low level flow "B" in Figure 2, U.S.G.D. 62360. crossed nicols, approx. 75 x. Note similarity of texture and composition to U.S.G.D. 62359 (Fig. 12) and U.S.G.D. 62354 (Fig. 13).

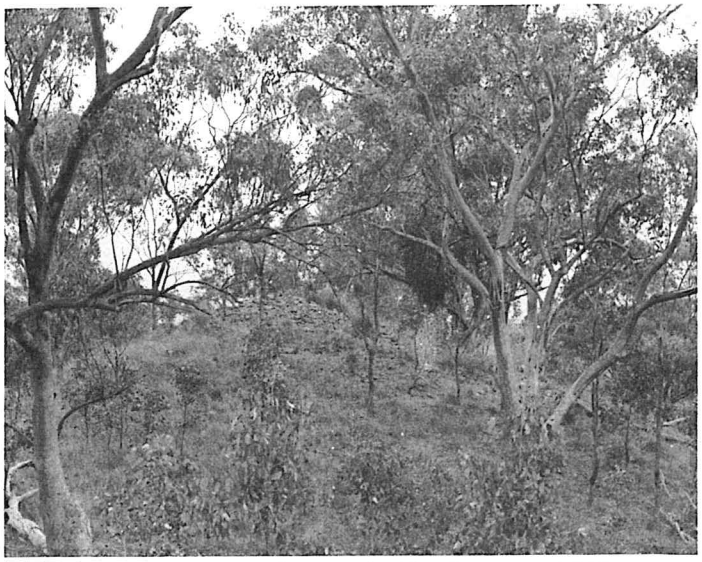


Figure16. Basalt knoll on south side of outcrop at "A" Fig. 2. Note conical shape and angular nature of basalt blocks at top of cone.



Figure 17. Tuff underlying basalt exposed in gully within the valley of upper Isaacs Creek (C in Fig 2). Further down hill tuff unconformably overlies the Yarrimie Formation.



Figure 18. View of Isaacs Creek valley from Caves Ridge looking east. Cuesta topography on southern valley side is developed on siliceous mudstones of the Yarrimie Formation.

"not a prominent feature in the landscape" in prebasaltic times. Connolly (1983) considered that Connolly and Francis' mapping of basalt at low levels upstream in Isaacs Creek was in error due to mass movement of weathered basalt.

The nature of the sub-basaltic (or prebasaltic) surface, like the age of the caves, is thus controversial. Considerable evidence has been canvassed by Connolly and Francis (1979) indicating that the sub-basaltic surface was irregular.

Connolly and Francis also argued that the basalt flows in the Isaacs Creek area were localised, identifying five basalt knolls ("G" and "H" in Fig. 2) associated with strong magnetic anomolies that they considered to be plugs, sources for localised flows. Connolly (1983) rejected this interpretation concluding instead that the knolls were "accumulations of corestones exposed by the removal of finer material". In addition to the five knolls identified by Connolly and Francis, a knoll is found at "I" in Fig 2 and one of Robson's low level basalts ("A" in Fig. 2) also has a knoll associated with it.

The basalt knolls do not consist of corestones, but rather of angular, relatively unweathered basaltic fragments in a cone-shaped outcrop (Fig. 16). This, along with the relatively coarse grainsize and presence of xenocrysts indicates that the knolls are most likely plugs as originally proposed.

An irregular sub-basaltic surface is supported by more recent observations.

Robson (1982) mapping the geology of the Yarrabin district (an area extending to within 3 km south west of Timor Caves) mapped basalt flows with bases as low as 510 m A.S.L. (Fig. 2). The closest of these flows to the caves ("B", Fig. 2) rests on the interfluve between the Isis River and Timor Creek and has a basal elevation of 560 m A.S.L., 110 m above the adjoining bed of the Isis River.

To the east of the caves where Connolly and Francis (1979) mapped the base of the basalt down to 660 m A.S.L. field investigation has found unambiguous basalt outcrops down to 730 m while adjoining higher hills are free from basalt. Basalt-free hills in the caves area with elevations over 700 m A.S.L. are shown in Fig. 2.

East of Timor Gap basalt extends down to an elevation of 830 m (D in Fig. 2) where it is well within the valley sides.

In the area mapped by Connolly and Francis the basalt is underlain by tuff forming the base of the Liverpool Range Beds. Slumps are common here with mass movement being due to slippage between the Lower Palaeozoic basement and the basalt when the tuff becomes mobilized. In places well within the valleys, however, it is clear that the tuff and the overlying basalt are in-situ (Fig. 17) and overlie sloping valley walls formed of Lower Palaeozoic, rocks indicating that valley development predated the basalt.

Taken together these observations indicate that the relief on the base of the basalt within 5 km of the caves must be at least 150 m (a similar figure to that found by Robson (1982) in the Yarrabin district) and that significant valley development took place prior to extrusion of the basalt.

Interpretation of the basalt body in Belfry Cave as a flow indicates that significant relief existed at Timor Caves at an even earlier time. The basalt shows no sign of interaction with standing water during its emplacement, thus it must have entered an air-filled cave. For this to occur the karst water table must have had an elevation lower than 550 m , i.e. less than 30 m above the present creek bed, in the Late Cretaceous making the landscape then, and the elevation of Caves Ridge above Isaacs Creek, much the same as we see today .

Origin of the Present Landscape

In addition to the evidence for an irregular prebasaltic surface the character of the present landscape is difficult to reconcile with development *ab initio* on a relatively featureless basaltic surface.

The Isis River, the main drainage feature of the area, follows the strike of Lower Palaeozoic rocks while its tributaries such as Farm Gully, Seckolds Gully, and probably the lower tract of Isaacs Creek are developed along cross faults. If the present topography were largely the result of erosion of a relatively flat basalt-covered landscape it is likely that the present drainage would reflect the palaeogeography of the basalt surface, rather than the structure of the Palaeozoic basement.

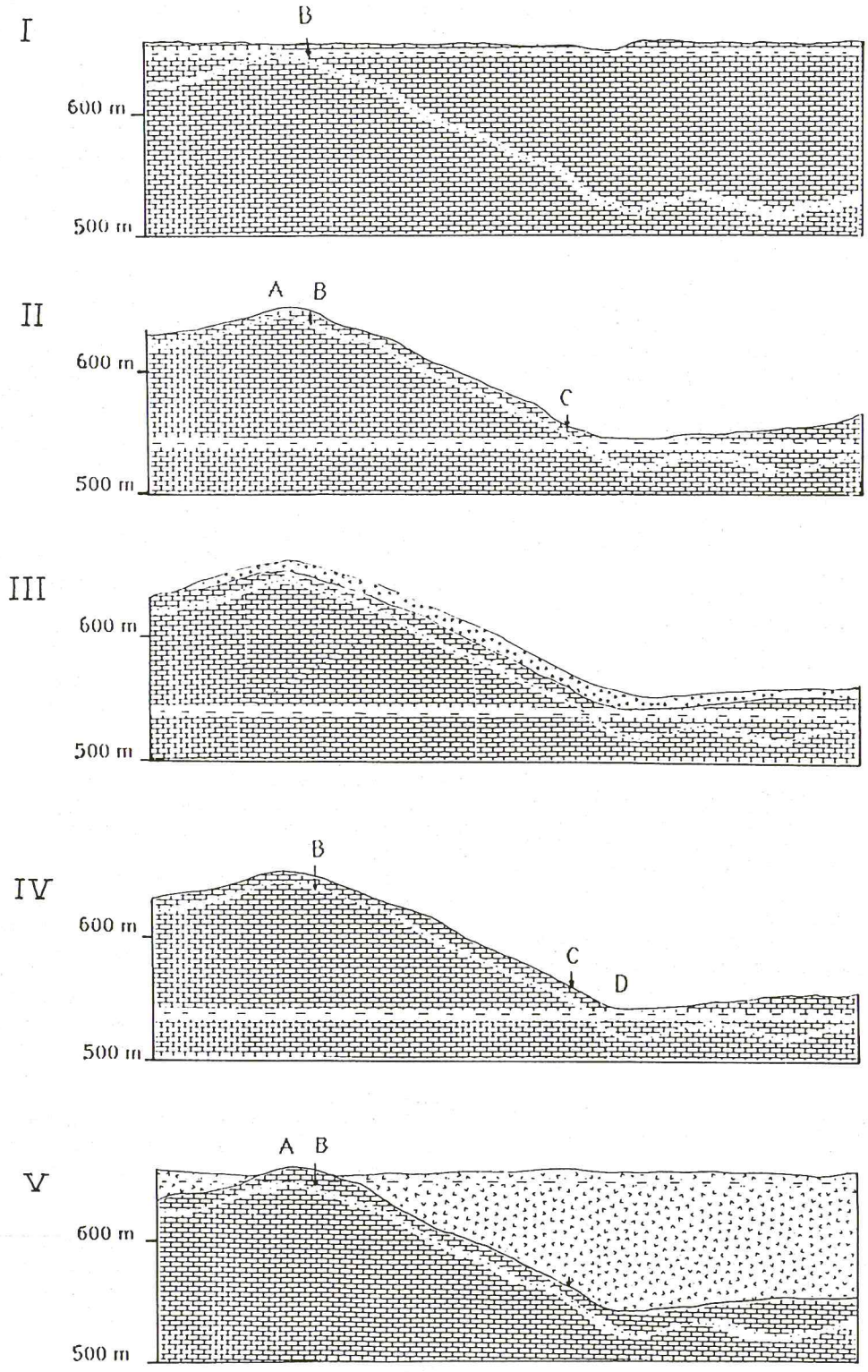
The development of drainage controlled by basalt palaeogeography has been shown to occur in the Barrington Tops area by Pain (1983). This has resulted in a radial drainage pattern, quite unlike the trellis pattern found in the Isaacs Creek area.

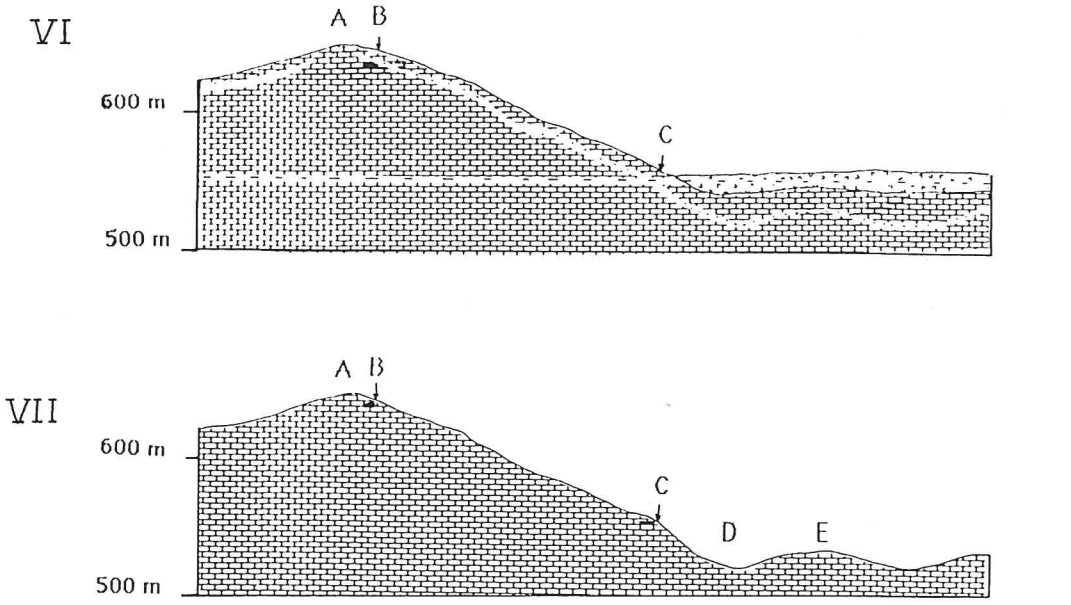
Another feature which demonstrates the significance of bedrock rather than basalt control over the landscape is the development of cuestas, dip slope hills, which are a prominent feature of the Isaacs Creek landscape (Fig. 18).

The drainage pattern and the cuestas further support the idea of the Isaacs Creek area having inherited much of its relief from prebasaltic times.

HISTORY OF SPELEOGENESIS

Cave development in the area now occupied by Main Cave preceded extrusion of Late Cretaceous basalt (Fig. 19, I). The development of extensive vadose palaeokarst indicates that prior to latest Cretaceous times the area now occupied by Main Cave was well above the water table (Fig. 19, II).


I
B
600 m
500 m
II
A
B
C
600 m
500 m
III
600 m
500 m
IV
B
C
D
600 m
500 m
V
A
B
600 m
500 m



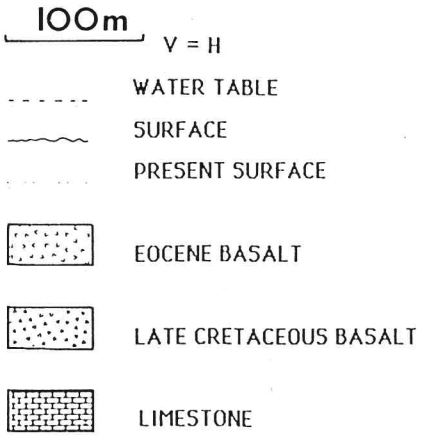


Figure. 19 Cartoon showing cave and landscape development at Timor Caves, based on cross-section through Main and Belfry Caves (see Fig. 3). A, Caves Ridge; B, Main Cave; C, Belfry Cave; D, Isaacs Creek; E, Road.

I Prior to incision of Isaacs Creek. Phreatic cave forms in position of present Main Cave. Water table >640 m A.S.L.

II Pre Late Cretaceous. Water table below level of present Belfry Cave. Vadose speleothem deposited in cave where Main Cave now located. (Corresponds to "A 1" in Fig. 9)

III Late Cretaceous. Basalt flow enters vadose caves forming bodies now exposed in Main and Belfry Caves. (Corresponds to "A 2" in Fig. 9)

IV Late Cretaceous to Eocene. Basalt flow is eroded off surface of limestone. Basalt remains as fill in caves.

V Eocene. Basalt and tuff of the Liverpool Range Beds fills Isaacs Creek Valley. This raises the water table to >640 m A.S.L. Main Cave is formed by phreatic solution.

VI Post Eocene. Basalt is eroded from Isaacs Creek. Belfry Cave becomes a stream sink when approx 30m of basalt still remains in the valley.

VII Present condition with Main and Belfry Caves above the water table.

If the Late Cretaceous water table was below the level of Belfry Cave, as proposed above, then phreatic solution of palaeo-Main Cave occurred much earlier, prior to the excavation of Isaacs Creek which must have been almost complete by Late Cretaceous times.

The small size of the basalt infilling compared to the extensive nature of the palaeokarst suggests that perhaps only a small vadose slot was still open in the Main Cave area, the original larger cave having almost completely been filled with speleothems.

In Eocene times basalt filled the valley of Isaacs Creek. This raised the water table and kept it high and stable long enough to cause nothephreatic solution of bedrock and palaeokarst resulting in the formation of Main Cave (Fig. 19, V). Belfry Cave and other caves at the same level, e.g. Shaft Cave and Helictite Cave, were formed at a later time when headward erosion had removed some of the basalt fill (Fig. 19, VI).

The development of phreatic roof pendants in flowstone in Belfry Cave seems to be related to a local backflooding event rather than to a major change in water table.

The detailed history of Belfry Cave and the later history of Main Cave cannot be elucidated without further study. Both contain thick deposits of entrance facies while Belfry Cave also contains much laminated clay. A study of these, and of the other caves in the area is essential if the problems of of cave and landscape development in the area are to be finally resolved.

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REFERENCES

ANDREWS, E.C., 1911. Geographical unity of eastern Australia in the late and post Tertiary times with applications to biological problems. *J. Proc. Roy. Soc. N.S.W.* *44*(4): 420-480.

BENSON, W.N., 1913. The geology and petrology of the Great Serpentine Belt of New South Wales, Part I. *Proc. Linn. Soc. N.S.W. 38*: 490-517.

CONNOLLY, M.D., 1983. Reassessment of cave ages at Isaacs Creek. *Helictite 21* (2): 64-65.

CONNOLLY, M.D., & FRANCIS, G., 1979. Cave and landscape evolution at Isaacs Creek, New South Wales. *Helictite 17*(1): 5-25.

CROOK, K.A.W., 1961a. Stratigraphy of the Tamworth Group (Lower and Middle Devonian), Tamworth-Nundle District, N.S.W. *J. Proc. Roy. Soc. N.S.W. 94*: 173-188.

CROOK, K.A.W.,1961b. Post-Carboniferous stratigraphy of the Tamworth-Nundle District, N.S.W. *J. Proc. Roy. Soc. N.S.W. 94*: 209-213.

DULHUNTY, J.A., 1976. Potassium-argon ages of igneous rocks in the Wollar-Rylstone region, New South Wales. *J. Proc. Roy. Soc. N.S.W. 109*: 35-39.

FRANCIS, G., & OSBORNE, R.A.L., 1983. Comment - Reassessment of cave ages at Isaacs Creek. *Helictite 21* (2): 66.

GALLOWAY, R.W., 1967. Pre-basalt, sub-basalt and post-basalt surfaces of the Hunter Valley, New South Wales. *in* J.N. Jennings and J.A. Mabbutt eds. LANDFORM STUDIES FROM AUSTRALIA AND NEW GUINEA. A.N.U. Press, Canberra : 293-314.

HOLLIS, J.D., & SUTHERLAND F.L., 1985. Occurrences and origins of gem zircons in eastern Australia. *Rec. Aust. Mus. 36*: 299-311.

JAMES, J., MIDDLETON, G., MONTGOMERY, N., PARKER, F., ROLLS, D., SCOTT, P., & WELLINGS, G., 1976. Timor Caves. *Sydney Speleological Society Occasional Paper. 6*: 1-50.

McDOUGALL, I., & WILKINSON, J.F.G.., 1967. Potassium-argon dates on some Cainozoic volcanic rocks from northeastern New South Wales. *J. geol. Soc. Aust. 14*(2): 225-223.

OFFENBERG, A.C., comp. 1971. Tamworth 1: 250,000 Geological Sheet. Department of Mines, Sydney.

OSBORNE, G.D., JOPLING, A.V., & LANCASTER, H.E., 1948. The stratigraphy and general form of the Timor Anticline, N.S.W. *J. Proc. Roy. Soc. N.S.W. 82* : 312-318.

OSBORNE, R.A.L., 1984a. Multiple karstification in the Lachlan Fold Belt in New South Wales: Reconnaissance evidence. *J. Proc. Roy. Soc. 117* : 15-34.

OSBORNE, R.A.L., 1984b. Lateral facies changes, unconformities and stratigraphic reversals: Their significance for cave sediment stratigraphy. *Trans. Brit. Cave Res. Assn. 11* (3) : 175-184.

PAIN, C.F., 1983. Geomorphology of the Barrington Tops area, New South Wales. *J. geol. Soc. Aust. 30* (2) : 187-194.

POGSON, D.J., 1972 Geological Map of New South Wales, Scale 1:1,000,000. Geol. Surv. N.S.W., Sydney.

ROBSON, A.D., 1982. Sedimentology and stratigraphy of middle Palaeozoic submarine fan deposits of the Yarrabin district, N.S.W. *BSc (Hons) Thesis, University of Sydney.* unpl.

SCHÖN, R.W., 1982. Petrology of volcanic rocks from the Liverpool Range, N.S.W. Abs. *Geological Society of Australia Abstracts 7*: 14.

WARDEN, D.E., ed. 1981. Timor Caves descriptions (Isis River). *Hills Speleology Club Yearbook 1980-81* : 9-30.

WELLMAN, P., & McDOUGALL, I., 1974. Potassium-Argon ages on the Cainozoic volcanic rocks of New South Wales. *J. geol. Soc. Aust. 21* (3) : 247-272.

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